Balancing chemical equations

You do not need to draw molecular diagrams each time to balance chemical equations. Look at the example below.

sodium chloride + fluorine gas ⟶ sodium fluoride + chlorine gas

$NaCl + F_2 \longrightarrow NaF + Cl_2$

At the start there is

1 atom Na + 1 atom Cl + 2 atoms F = 4 atoms in total

At the end there is

1 atom Na + 2 atoms Cl + 1 atom F = 4 atoms in total

Even though the total number of atoms is the same, the equation is not balanced because we have more atoms of fluorine and less of chlorine at the start of the reaction. We use trial and error to see what numbers we need to add to the front of the compounds.

$\mathbf{2}NaCl + F_2 \longrightarrow \mathbf{2}NaF + Cl_2$

If we put a 2 in front of the NaCl we now have 2 atoms of chlorine each side but now we have 2 atoms of sodium at the start of the reaction, so we also have to put a two at the front of NaF. Now there are two fluorine atoms on each side, so the equation is balanced.

Remember: A large number at the front means you multiply the number of atoms in one molecule by that number.

☞ **1** Count the number of atoms in the following. The first one has been done.

a $2Na_2SO_4$ 4 atoms of sodium, 2 atoms of sulfur, 8 atoms of oxygen

b $3CH_4$ _____

c $2H_2O$ _____

d $2CaCO_3$ _____

2 Now try balancing the following equations in your workbook.

a __K + __ $MgBr_2$ ⟶ __ KBr + __Mg

b __H_2O_2 ⟶ __H_2O + __ O_2

c __ NO_2 ⟶ __O_2 + __ N_2

d __ N_2 + __ H_2 ⟶ __ NH_3

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